

```
1  #include <stdio.h>
2  #include <stdlib.h>
3  int mutex = 1;
4  int full = 0;
5  int empty = 3, x = 0;
6  int wait(int s) {
7      return (--s);
8  }
9  int signal(int s) {
10     return (++s);
11 }
12 void producer() {
13     mutex = wait(mutex);
14     full = signal(full);
15     empty = wait(empty);
16     x++;
17     printf("\nProducer produces item %d", x);
18     mutex = signal(mutex);
19 }
20 void consumer() {
21     mutex = wait(mutex);
22     full = wait(full);
23     empty = signal(empty);
24     printf("\nConsumer consumes item %d", x);
25     x--;
26     mutex = signal(mutex);
27 }
```



```
36         if((mutex == 1) && (empty != 0))
37             producer();
38         else
39             printf("Buffer is Full");
40         break;
41     case 2:
42         if((mutex == 1) && (full != 0))
43             consumer();
44         else
45             printf("Buffer is Empty");
46         break;
47
48     case 3:
49         exit(0);
50     }
51 }
52 return 0;
53 }
```

1. Producer
2. Consumer
3. Exit

Enter your choice: 1

Producer produces item 1

Enter your choice: 1

Producer produces item 2

Enter your choice: 2

Consumer consumes item 2

Enter your choice: 2

Consumer consumes item 1

Enter your choice: 3

Process exited after 33.46 seconds with return value 0

Press any key to continue . . . |